## APPENDIX II

## BURUNDI: DEBT SUSTAINABILITY ANALYSIS USING THE LOW-INCOME COUNTRY FRAMEWORK

1. **This debt sustainability analysis (LIC DSA) assesses Burundi's external and public debt dynamics using the forward-looking debt sustainability framework (DSF) for low-income countries**.[1] The LIC DSA is based on the reconciled debt database used for the HIPC DSA and incorporates the impact of HIPC and MDRI relief in the baseline. Three key methodological differences explain variations in the LIC DSA results from those of the HIPC DSA: (i) the discount rate is fixed at 5 percent, compared to the currency-specific 6-month average of commercial interest reference rate (CIRR) used in the HIPC DSA; (ii) the export denominator is based on the current level of exports, rather than the three-year backward average in the HIPC DSA; and (iii) exchange rate projections are used to convert the debt data after debt relief. [2]

2. **Despite the delivery of HIPC and MDRI assistance, Burundi has a high risk of debt distress.** Nevertheless, this is an improvement compared to the previous LIC DSA, which found Burundi to be "in debt distress".[3] After full delivery of HIPC assistance, bilateral debt relief beyond the HIPC Initiative and MDRI, Burundi's present value of debt-to-exports ratio is expected to remain above the country-specific policy dependent threshold for most of the projection period. Under the standard alternative scenarios and stress tests, all indicators with the exception of the debt-to-revenue and debt service-to-revenue ratios exceed the country-specific debt burden thresholds. Given these results, Burundi should continue to rely on grants and highly concessional loans to meet its financing needs, while intensifying efforts to increase and diversify its export base.

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[1] See "Debt Sustainability in Low-Income Countries: Proposal for an Operational Framework and Policy Implications" (http://www.imf.org/external/np/pdr/sustain/2004/020304.htm and IDA/SECM2004/0035, 2/3/04) and "Debt Sustainability in Low-Income Countries: Further Considerations on an Operational Framework and Policy Implications" (http://www.imf.org/external/np/pdr/sustain/2004/091004.htm and IDA/SECM2004/0629, 9/10/04).

[2] The present value of debt is calculated by discounting debt service converted in dollar terms using the WEO average exchange rate with respect to the U.S. dollar for the first 5 years of the projection period, and maintaining the exchange rates constant thereafter. This differs from the HIPC methodology, where the present value of debt is calculated by applying currency specific discount rates to debt service converted into dollar at fixed end-2007 exchange rates. See "Staff Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries" October 6, 2008.

[3] This appendix updates the DSA prepared in December 2007 (IMF Country Report No. 08/27, January 23, 2008).

3. **Four key differences underpin the results of this LIC DSA relative to the previous one**: (i) debt relief is based on the calculations of the HIPC completion point rather than on earlier preliminary estimates; (ii) real GDP growth is assumed to be lower by about 1 percentage point, closer to the historical average; (iii) the baseline scenario assumes lower growth of exports and imports of goods and services, partly stemming from lower WEO prices for primary commodities and oil respectively; and (iv) nominal GDP is higher due to the spike in inflation as a result of higher food and oil prices in 2007 and the first half of 2008.

## RECENT DEVELOPMENTS

4. **Burundi's public and publicly guaranteed (PPG) external debt decreased slightly in 2007.** Nominal PPG debt amounted to US$1.5 billion or 155 percent of GDP in 2007 while the present value (PV) of debt-to-exports ratio stood at 104.8 percent. About 86 percent of Burundi's outstanding external PPG debt was owed to multilateral creditors, with bilateral creditors accounting for the remainder. External debt service in 2007 was reduced to 6.8 percent of exports in 2007 from 10.7 percent in 2006, reflecting primarily debt relief by Paris Club creditors.

Table 1. Burundi: Public External Debt, 2004-07

| | 2004 | | 2006 | | 2007 | |
|---|---|---|---|---|---|---|
| | U.S. Dollars | Percent of Total | U.S. Dollars | Percent of Total | U.S. Dollars | Percent of Total |
| Total | 1,384 | 100.0 | 1,472 | 100.0 | 1,468.0 | 100.0 |
| Multilateral | 1170 | 84.5 | 1253 | 85.1 | 1,262 | 86.0 |
| Bilateral | 208 | 15.0 | 213 | 14.5 | 205.4 | 14.0 |
| Paris Club | 145.1 | 10.5 | 146.3 | 9.9 | 152.1 | 10.4 |
| Non-Paris Club | 62.5 | 4.5 | 67.0 | 4.6 | 53.3 | 3.6 |
| Commercial | 6.6 | 0.5 | 5.8 | 0.4 | 0.6 | 0.0 |

Source: Burundi authorities; and staff estimates.

## KEY ASSUMPTIONS

5. **The baseline macroeconomic outlook has been updated to reflect recent developments (Box 1)**. The near term outlook will be influenced by the adverse impact of the international financial crisis. Provided the security situation continues to improve, Burundi's medium and long-term outlook is broadly positive. Real GDP growth is expected to average 5 percent over the long term, sustained by the consolidation of peace, maintenance of macroeconomic stability, continued improvements in the business environment, and greater openness to trade, following Burundi's accession to the East African Community (EAC). Investment is expected to average about 16 percent of GDP, a level considered to be supportive of economic growth. Exports of goods and services are assumed to drive growth over the long term, following investments in key export sectors over the medium term (traditional and non-traditional). The external current account deficit, including grants, will decline over time, despite lower donor support. Financing

will continue to be in the form of grants and highly concessional loans. Fiscal policy would be supportive of economic growth and poverty reduction, while preserving debt sustainability. Revenue mobilization efforts would lead to a gradual increase in government revenues, while expenditure would gradually decline to more sustainable levels, once the current transition phase ends.

6. **Risks to the macroeconomic outlook are significant.** A worsening of the political, social and security situation would endanger donor support and lead to a rapid deterioration of debt indicators. Exogenous shocks, including a stronger-than-expected impact of the international economic crisis on exports and growth and a reversal of the current downward trend in oil prices, would lead to a further deterioration in debt indicators.

## EXTERNAL DEBT SUSTAINABILITY ANALYSIS

7. **All but one debt indicators are expected to remain below the country-specific debt burden thresholds under the baseline scenario.** Following HIPC and MDRI relief, the present value (PV) of debt-to-GDP, the PV of debt-to-revenues, the debt service-to exports ratio, and the debt service-to revenue ratios are expected to remain well below the indicative country-specific thresholds throughout the projection period (Box 2 and Table 1).[4] All debt and debt-service indicators increase modestly over the medium term and then gradually level off. In contrast, the PV of debt-to-exports ratio will remain above the 100 percent threshold until 2019 and then decline rapidly to about 70 percent at the end of the projection period. The evolution of the present value of debt-to-exports ratio reflects the country's development needs over the medium term and the gradual improvement in export capacity.

[4] The latest World Bank Country Policy and Institutional Assessment (CPIA) rates Burundi as a poor performer. The three-year moving average CPIA in 2007 is 3.0. This score has remained unchanged during the last three years. A rating below 3.25 corresponds to a poor policy performance, according to the LIC DSF.

### Box 1: Main Macroeconomic Assumptions, 2008–28

**Real GDP growth** is expected to accelerate to 5 percent over the medium term reflecting the consolidation of peace, the reform of the coffee sector, and economic stimulus from increased capital spending associated with donor support. Long-term growth is expected to be about 5 percent. Investment is expected to average about 16 percent of GDP, a level considered to be supportive of long-term economic growth.

**Inflation** is assumed to decline gradually over the medium term and stabilize at about 4 percent over the long term, as the central bank moves to strengthen its policy instruments.

**Fiscal policy** would be supportive of economic growth and poverty reduction, while maintaining debt sustainability and macroeconomic stability.

- **Government revenues**, excluding grants, are projected to remain rather stable over the medium term, but gradually increase to about 21 percent of GDP over the long term. This increase reflects primarily a widening of the tax base as a result of continued tax administration reforms.
- **Government expenditure** is expected to gradually decline over time to the more sustainable level of about 27 percent of GDP. This path is consistent with a gradual reduction of current spending growth, and higher pro-poor spending. Public investment is expected to gradually decline over time to about 4 percent of GDP as the private sector becomes the engine of growth.

**Exports of goods and services** are projected to increase by an average of 7 percent over the medium term reflecting in part lower commodity prices; they are projected to increase by 9–10 percent over the longer term as investment in key export sectors (traditional and non-traditional) begins to yield results. In volume terms, exports of goods are projected to gradually increase.

**Imports of goods and services** will gradually decelerate over time to a more sustainable rate of 5 percent. With emergency and reconstruction-related imports winding down and oil prices declining over the medium term, merchandise import volumes in the outer years are expected to move broadly in line with real GDP growth.

**Current account deficit, including grants,** is expected to deteriorate to about 17 percent of GDP in 2015, and then gradually decline to about 9 percent at the end of the projection period. Excluding transfers, the current account deficit is assumed to decline from 26 percent of GDP to about 9 percent at the end of the projection period. The authorities will rely largely on highly concessional loans to finance the current account deficit.

**External financing** will increasingly rely on loans in the long term, as grants gradually subside. Grant support is assumed to decline from about 26 percent of GDP in 2008 to about 10 percent in 2015 as humanitarian assistance gradually declines. Loans are assumed to gradually decline from about 3 percent of GDP in 2008 to about 1 percent at the end of the projection period. Borrowing is assumed to take place primarily on highly concessional terms.

**Box 2. Burundi: Summary of Baseline Debt Sustainability Indicators**
(Percent)

| | Indicative Threshold | 2008 | 20018 | 2028 | Average 2008-28 |
|---|---|---|---|---|---|
| NPV of debt to GDP | 30 | 9.4 | 9.9 | 7.4 | 9.7 |
| NPV of debt to exports | 100 | 96.2 | 113.4 | 68.5 | 109.0 |
| NPV of debt to revenue | 200 | 49.3 | 47.9 | 35.3 | 48.1 |
| Debt service to exports | 15 | 3.0 | 8.7 | 4.2 | 6.1 |
| Debt service to revenues | 25 | 1.5 | 3.7 | 2.2 | 2.7 |

Sources: Burundi authorities; and staff estimates.

8. **Alternative scenarios and stress tests underscore the underlying risks**. Under the historical scenario, debt indicators are considerably worse over the medium term, but improve rapidly relative to the baseline scenario. The results of the historical scenario should be interpreted with caution, given Burundi's history of conflict. The scenario assuming new borrowing at a higher interest rate shows that debt indicators at the end of the projection period are generally worse than in the baseline scenario. This result underscores the need for continued reliance on grants and highly concessional financing, especially since the grace period and maturity under the alternative scenario are identical to the baseline scenario. Burundi is also vulnerable to a variety of shocks, most notably to a combination of growth, export, inflation, and FDI/current transfers shock. The bound tests suggest that in the presence of shocks, all debt indicators, with the exception of the debt service to revenue ratio will exceed the indicative debt-burden thresholds.

9. **The risk of debt distress will remain high following the HIPC completion point.** The preceding analysis suggests that there is a protracted breach of the present value of debt-to-exports threshold under the baseline scenario, while alternative scenarios and bound tests suggest that all debt burden thresholds are likely to be exceeded in the event of shocks.

## FISCAL DEBT SUSTAINABILITY

10. **Public debt indicators are expected to gradually improve under the baseline scenario.** Public external debt accounts for almost the entire stock of public debt.[5] The ratios of the present value of public debt-to-GDP and public debt-to-revenues are low reflecting Burundi's reliance on grants to finance the reconstruction and poverty reduction efforts. They are expected to decline over time as the public sector borrowing requirement declines reflecting the improvement in the revenue base and the gradual decline in government spending in the post-reconstruction period.

11. **The alternative scenarios and stress tests show that public debt indicators can worsen rapidly.** All debt indicators worsen assuming an unchanged primary balance from 2008, highlighting the need for fiscal prudence over time. A scenario based on the historical growth and primary balance also leads to a rapid deterioration of debt indicators, thus underscoring the importance of avoiding past unsustainable policies. Under the most extreme shock, which assumes lower real GDP growth and higher primary deficit, all debt indicators worsen rapidly. Nevertheless, the debt service-to-revenue ratio increases only gradually, as additional borrowing is on highly concessional terms.

## CONCLUSION

12. **Burundi will remain at a high risk of debt distress following the HIPC completion point and debt relief under MDRI.** Despite the positive impact of debt relief, the PV of external debt-to-exports will remain above the country-specific debt burden threshold over the long term. Burundi is also vulnerable to exogenous shocks which could lead to a considerable worsening of debt indicators. Burundi should therefore continue to implement prudent policies that support macroeconomic stability. In particular, it should continue to rely on grants and highly concessional loans to meet its financing needs, while intensifying efforts to improve tax administration and widen the tax base. Finally, it should diversify the export base and encourage private sector development by addressing existing weaknesses in the business environment.

[5] The baseline scenario assumes no issuance of medium-term domestic debt, as the market for government debt is not expected to play a significant role in meeting the government's financing needs in the foreseeable future.

Table 1: External Debt Sustainability Framework, Baseline Scenario, 2005-2028 1/
(In percent of GDP, unless otherwise indicated)

| | Actual | | | Historical Average | Standard Deviation | Projections | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2005 | 2006 | 2007 | | | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2008-2013 Average | 2018 | 2028 | 2014-2028 Average |
| **External debt (nominal) 1/** | **169.8** | **154.8** | **148.8** | | | **122.0** | **26.3** | **27.0** | **27.3** | **26.9** | **26.5** | | **22.7** | **16.0** | |
| o/w public and publicly guaranteed (PPG) | 169.8 | 154.8 | 148.8 | | | 122.0 | 26.3 | 27.0 | 27.3 | 26.9 | 26.5 | | 22.7 | 16.0 | |
| Change in external debt | -52.0 | -15.0 | -6.0 | | | -26.8 | -95.6 | 0.7 | 0.3 | -0.3 | -0.4 | | -1.4 | -0.5 | |
| Identified net debt-creating flows | -39.8 | -11.2 | -1.2 | | | -1.7 | -1.9 | 0.0 | 1.9 | 3.4 | 4.2 | | 9.2 | 5.8 | |
| **Non-interest current account deficit** | **-1.1** | **14.3** | **15.3** | **6.0** | **5.5** | **12.9** | **11.4** | **10.0** | **12.0** | **13.6** | **14.4** | 12.4 | **14.4** | **7.8** | 11.8 |
| Deficit in balance of goods and services | 29.1 | 38.5 | 39.8 | | | 36.7 | 28.6 | 26.2 | 26.1 | 25.7 | 25.2 | | 21.3 | 12.7 | |
| Exports | 11.5 | 10.1 | 8.6 | | | 9.8 | 7.6 | 8.1 | 8.2 | 8.2 | 8.3 | | 8.7 | 10.8 | |
| Imports | 40.6 | 48.6 | 48.3 | | | 46.5 | 36.2 | 34.3 | 34.4 | 33.9 | 33.5 | | 30.1 | 23.5 | |
| Net current transfers (negative = inflow) | -30.1 | -24.9 | -24.6 | -15.2 | 9.5 | -24.1 | -17.5 | -16.6 | -14.5 | -12.5 | -11.2 | | -7.1 | -6.2 | -7.0 |
| o/w official | -27.9 | -21.8 | -21.6 | | | -20.9 | -14.3 | -13.5 | -11.5 | -9.6 | -8.3 | | -5.5 | -4.7 | |
| Other current account flows (negative = net inflow) | -0.1 | 0.8 | 0.2 | | | 0.3 | 0.3 | 0.3 | 0.3 | 0.4 | 0.3 | | 0.2 | 1.2 | |
| **Net FDI (negative = inflow)** | **-3.4** | **-3.8** | **-8.1** | **-4.0** | **2.5** | **-9.0** | **-9.0** | **-8.9** | **-8.9** | **-9.0** | **-9.1** | | **-4.2** | **-1.3** | -3.4 |
| **Endogenous debt dynamics 2/** | **-35.4** | **-21.7** | **-8.5** | | | **-5.6** | **-4.4** | **-1.0** | **-1.1** | **-1.1** | **-1.1** | | **-1.0** | **-0.7** | |
| Contribution from nominal interest rate | 2.3 | 0.2 | 0.4 | | | 0.0 | 0.1 | 0.1 | 0.1 | 0.2 | 0.2 | | 0.2 | 0.1 | |
| Contribution from real GDP growth | -1.7 | -7.6 | -5.2 | | | -5.6 | -4.5 | -1.2 | -1.3 | -1.3 | -1.3 | | -1.1 | -0.8 | |
| Contribution from price and exchange rate changes | -36.0 | -14.3 | -3.6 | | | … | … | … | … | … | … | | … | … | |
| **Residual (3-4) 3/** | **-12.2** | **-3.9** | **-4.8** | | | **-25.1** | **-93.7** | **0.6** | **-1.6** | **-3.8** | **-4.6** | | **-10.5** | **-6.3** | |
| o/w exceptional financing | -4.9 | -4.3 | -5.9 | | | -7.8 | -84.3 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| PV of external debt 4/ | … | … | 9.0 | | | 9.4 | 9.9 | 11.0 | 12.3 | 12.2 | 12.1 | | 9.9 | 7.4 | |
| In percent of exports | … | … | 104.8 | | | 96.2 | 131.7 | 136.3 | 149.7 | 149.0 | 146.1 | | 113.4 | 68.5 | |
| **PV of PPG external debt** | **…** | **…** | **9.0** | | | **9.4** | **9.9** | **11.0** | **12.3** | **12.2** | **12.1** | | **9.9** | **7.4** | |
| **In percent of exports** | **…** | **…** | **104.8** | | | **96.2** | **131.7** | **136.3** | **149.7** | **149.0** | **146.1** | | **113.4** | **68.5** | |
| **In percent of government revenues** | **…** | **…** | **48.2** | | | **49.3** | **51.6** | **56.5** | **63.1** | **62.4** | **61.0** | | **47.9** | **35.3** | |
| **Debt service-to-exports ratio (in percent)** | **28.4** | **10.7** | **6.8** | | | **3.0** | **3.7** | **2.4** | **4.0** | **7.9** | **8.8** | | **8.7** | **4.2** | |
| **PPG debt service-to-exports ratio (in percent)** | **28.4** | **10.7** | **6.8** | | | **3.0** | **3.7** | **2.4** | **4.0** | **7.9** | **8.8** | | **8.7** | **4.2** | |
| **PPG debt service-to-revenue ratio (in percent)** | **16.3** | **5.7** | **3.1** | | | **1.5** | **1.4** | **1.0** | **1.7** | **3.3** | **3.7** | | **3.7** | **2.2** | |
| Total gross financing need (Millions of U.S. dollars) | -9.5 | 106.8 | 76.3 | | | 48.8 | 38.8 | 20.4 | 56.8 | 94.3 | 115.8 | | 301.0 | 396.5 | |
| Non-interest current account deficit that stabilizes debt ratio | 50.9 | 29.4 | 21.4 | | | 39.7 | 107.1 | 9.4 | 11.7 | 13.9 | 14.8 | | 15.8 | 8.2 | |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 0.9 | 5.1 | 3.6 | 2.3 | 2.6 | 4.5 | 4.5 | 5.0 | 5.0 | 5.0 | 5.0 | 4.8 | 5.0 | 5.0 | 5.0 |
| GDP deflator in US dollar terms (change in percent) | 19.4 | 9.2 | 2.4 | -1.7 | 10.6 | 14.8 | 15.8 | 7.2 | 1.3 | 1.7 | 1.7 | 7.1 | 2.7 | 2.4 | 2.4 |
| Effective interest rate (percent) 5/ | 1.3 | 0.1 | 0.3 | 0.8 | 0.6 | 0.0 | 0.1 | 0.5 | 0.6 | 0.6 | 0.6 | 0.4 | 0.7 | 0.7 | 0.7 |
| Growth of exports of G&S (US dollar terms, in percent) | 44.8 | 1.0 | -10.2 | 1.0 | 24.0 | 36.8 | -6.5 | 20.0 | 8.4 | 6.7 | 7.6 | 12.1 | 9.5 | 8.8 | 9.5 |
| Growth of imports of G&S (US dollar terms, in percent) | 44.2 | 37.5 | 5.5 | 15.6 | 21.7 | 15.3 | -5.7 | 6.6 | 6.6 | 5.5 | 5.4 | 5.6 | 5.0 | 4.9 | 5.0 |
| Grant element of new public sector borrowing (in percent) | … | … | … | … | … | 52.5 | 48.7 | 48.6 | 48.5 | 58.8 | 58.8 | 52.6 | 58.8 | 58.8 | 58.8 |
| Government revenues (excluding grants, in percent of GDP) | 20.0 | 18.9 | 18.6 | | | 19.1 | 19.3 | 19.4 | 19.5 | 19.6 | 19.8 | | 20.7 | 21.0 | 20.8 |
| Aid flows (in Millions of US dollars) 7/ | 163.1 | 198.5 | 227.6 | | | 337.9 | 1224.5 | 363.5 | 344.1 | 334.9 | 352.7 | | 251.8 | 297.0 | |
| o/w Grants | 93.8 | 164.2 | 203.7 | | | 298.5 | 1185.5 | 324.9 | 305.9 | 300.5 | 316.0 | | 224.2 | 239.8 | |
| o/w Concessional loans | 69.3 | 34.3 | 24.0 | | | 39.4 | 39.0 | 38.6 | 38.2 | 34.4 | 36.7 | | 27.6 | 57.2 | |
| Grant-equivalent financing (in percent of GDP) 8/ | … | … | … | | | 27.8 | 85.8 | 22.2 | 19.7 | 17.7 | 17.5 | | 8.7 | 4.8 | 7.5 |
| Grant-equivalent financing (in percent of external financing) 8/ | … | … | … | | | 93.2 | 97.6 | 92.1 | 91.8 | 95.8 | 95.7 | | 95.5 | 92.1 | 93.5 |
| *Memorandum items:* | | | | | | | | | | | | | | | |
| Nominal GDP (Millions of US dollars) | 800.6 | 919.0 | 974.7 | | | 1169.7 | 1415.8 | 1592.5 | 1694.3 | 1808.2 | 1930.7 | | 2754.7 | 5718.0 | |
| Nominal dollar GDP growth | 20.5 | 14.8 | 6.1 | | | 20.0 | 21.0 | 12.5 | 6.4 | 6.7 | 6.8 | 12.2 | 7.8 | 7.5 | 7.5 |
| PV of PPG external debt (in Millions of US dollars) | | | 87.6 | | | 110.0 | 140.7 | 174.8 | 208.1 | 220.9 | 233.1 | | 273.3 | 424.6 | |
| (PVt-PVt-1)/GDPt-1 (in percent) | | | | | | 2.3 | 2.6 | 2.4 | 2.1 | 0.8 | 0.7 | 1.8 | 0.2 | 0.3 | 0.4 |

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2a.Burundi: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2008-2028
(In percent)

| | | | | | | | | | | | Projections | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | **2018** | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 |
| | | | | | | | | | | | PV of debt-to GDP ratio | | | | | | | | | | |
| **Baseline** | 9 | 10 | 11 | 12 | 12 | 12 | 12 | 11 | 11 | 11 | **10** | 9 | 9 | 9 | 9 | 8 | 8 | 8 | 8 | 8 | 7 |
| **A. Alternative Scenarios** | | | | | | | | | | | | | | | | | | | | | |
| A1. Key variables at their historical averages in 2008-2028 1/ | 9 | 18 | 21 | 23 | 23 | 23 | 20 | 17 | 13 | 10 | **6** | 1 | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| A2. New public sector loans on less favorable terms in 2008-2028 2/ | 9 | 10 | 11 | 12 | 12 | 12 | 12 | 11 | 11 | 11 | **11** | 10 | 10 | 10 | 10 | 10 | 9 | 9 | 9 | 9 | 9 |
| **B. Bound Tests** | | | | | | | | | | | | | | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2009-2010 | 9 | 10 | 12 | 14 | 13 | 13 | 13 | 13 | 12 | 12 | **11** | 10 | 10 | 10 | 9 | 9 | 9 | 9 | 9 | 8 | 8 |
| B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/ | 9 | 11 | 13 | 14 | 14 | 14 | 14 | 13 | 13 | 12 | **12** | 11 | 11 | 10 | 10 | 10 | 9 | 9 | 9 | 8 | 8 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010 | 9 | 13 | 18 | 20 | 20 | 19 | 19 | 18 | 18 | 17 | **16** | 15 | 15 | 14 | 14 | 14 | 13 | 13 | 13 | 12 | 12 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/ | 9 | 18 | 27 | 28 | 27 | 27 | 26 | 25 | 24 | 24 | **23** | 21 | 20 | 19 | 18 | 17 | 16 | 15 | 14 | 13 | 13 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 9 | 22 | 40 | 42 | 41 | 40 | 39 | 38 | 37 | 36 | **34** | 32 | 30 | 28 | 27 | 25 | 24 | 23 | 22 | 20 | 19 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/ | 9 | 13 | 15 | 17 | 16 | 16 | 16 | 15 | 15 | 14 | **13** | 13 | 12 | 12 | 12 | 11 | 11 | 11 | 11 | 10 | 10 |
| | | | | | | | | | | | PV of debt-to-exports ratio | | | | | | | | | | |
| **Baseline** | 96 | 132 | 136 | 150 | 149 | 146 | 141 | 134 | 128 | 122 | **113** | 105 | 98 | 93 | 88 | 84 | 81 | 78 | 74 | 71 | 69 |
| **A. Alternative Scenarios** | | | | | | | | | | | | | | | | | | | | | |
| A1. Key variables at their historical averages in 2008-2028 1/ | 96 | 244 | 265 | 286 | 283 | 273 | 245 | 201 | 155 | 111 | **64** | 14 | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| A2. New public sector loans on less favorable terms in 2008-2028 2/ | 96 | 131 | 134 | 146 | 146 | 144 | 140 | 136 | 132 | 131 | **124** | 117 | 111 | 105 | 100 | 96 | 92 | 89 | 86 | 83 | 80 |
| **B. Bound Tests** | | | | | | | | | | | | | | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2009-2010 | 96 | 132 | 136 | 150 | 149 | 146 | 141 | 134 | 128 | 122 | **113** | 105 | 98 | 93 | 88 | 84 | 81 | 78 | 74 | 71 | 68 |
| B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/ | 96 | 169 | 310 | 333 | 330 | 323 | 312 | 297 | 283 | 272 | **253** | 234 | 217 | 204 | 192 | 181 | 173 | 165 | 157 | 149 | 142 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010 | 96 | 132 | 136 | 150 | 149 | 146 | 141 | 134 | 128 | 122 | **113** | 105 | 98 | 93 | 88 | 84 | 81 | 78 | 74 | 71 | 68 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/ | 96 | 244 | 333 | 339 | 334 | 325 | 313 | 300 | 287 | 275 | **258** | 239 | 218 | 199 | 183 | 168 | 156 | 145 | 135 | 126 | 117 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 96 | 240 | 461 | 469 | 462 | 450 | 434 | 415 | 397 | 380 | **358** | 331 | 302 | 276 | 254 | 233 | 217 | 202 | 188 | 175 | 164 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/ | 96 | 132 | 136 | 150 | 149 | 146 | 141 | 134 | 128 | 122 | **113** | 105 | 98 | 93 | 88 | 84 | 81 | 78 | 74 | 71 | 68 |
| | | | | | | | | | | | PV of debt-to-revenue ratio | | | | | | | | | | |
| **Baseline** | 49 | 52 | 56 | 63 | 62 | 61 | 59 | 56 | 53 | 51 | **48** | 45 | 43 | 42 | 41 | 40 | 39 | 38 | 37 | 36 | 35 |
| **A. Alternative Scenarios** | | | | | | | | | | | | | | | | | | | | | |
| A1. Key variables at their historical averages in 2008-2028 1/ | 49 | 96 | 110 | 120 | 119 | 114 | 102 | 84 | 65 | 46 | **27** | 6 | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| A2. New public sector loans on less favorable terms in 2008-2028 2/ | 49 | 51 | 56 | 61 | 61 | 60 | 59 | 57 | 55 | 55 | **52** | 50 | 49 | 48 | 47 | 46 | 45 | 44 | 43 | 42 | 41 |
| **B. Bound Tests** | | | | | | | | | | | | | | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2009-2010 | 49 | 54 | 62 | 70 | 69 | 67 | 65 | 62 | 59 | 57 | **53** | 50 | 48 | 46 | 45 | 44 | 43 | 42 | 41 | 40 | 39 |
| B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/ | 49 | 55 | 68 | 74 | 73 | 71 | 69 | 65 | 62 | 60 | **56** | 53 | 51 | 49 | 47 | 46 | 44 | 43 | 42 | 40 | 39 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010 | 49 | 68 | 91 | 102 | 101 | 98 | 95 | 90 | 86 | 83 | **77** | 73 | 70 | 68 | 66 | 65 | 63 | 62 | 60 | 59 | 57 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/ | 49 | 96 | 138 | 143 | 140 | 136 | 131 | 125 | 120 | 115 | **109** | 103 | 96 | 90 | 85 | 80 | 76 | 72 | 68 | 64 | 60 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 49 | 115 | 208 | 216 | 211 | 205 | 197 | 189 | 180 | 173 | **165** | 155 | 145 | 136 | 129 | 122 | 115 | 109 | 103 | 97 | 92 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/ | 49 | 70 | 76 | 85 | 84 | 82 | 79 | 75 | 72 | 69 | **65** | 61 | 58 | 57 | 55 | 54 | 53 | 52 | 50 | 49 | 48 |

Table 2b.Burundi: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2008-2028
(In percent)

| | | | | | | | | | | | Projections | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2014 | 2015 | 2016 | 2017 | **2018** | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 |
| | | | | | | | | | | | Debt service-to-exports ratio | | | | | | | | | | |
| **Baseline** | 3 | 4 | 2 | 4 | 8 | 9 | 11 | 12 | 11 | 10 | **9** | 8 | 6 | 5 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| **A. Alternative Scenarios** | | | | | | | | | | | | | | | | | | | | | |
| A1. Key variables at their historical averages in 2008-2028 1/ | 3 | 6 | 5 | 7 | 13 | 15 | 19 | 21 | 21 | 19 | **17** | 20 | 17 | 13 | 9 | 8 | 7 | 5 | 3 | 1 | 0 |
| A2. New public sector loans on less favorable terms in 2008-2028 2/ | 3 | 5 | 3 | 5 | 10 | 11 | 12 | 12 | 10 | 7 | **6** | 6 | 6 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| **B. Bound Tests** | | | | | | | | | | | | | | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2009-2010 | 3 | 5 | 2 | 4 | 8 | 9 | 11 | 12 | 11 | 10 | **9** | 8 | 6 | 5 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/ | 3 | 6 | 5 | 8 | 16 | 17 | 21 | 23 | 22 | 19 | **17** | 15 | 14 | 11 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010 | 3 | 5 | 2 | 4 | 8 | 9 | 11 | 12 | 11 | 10 | **9** | 8 | 6 | 5 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/ | 3 | 5 | 4 | 7 | 11 | 12 | 13 | 14 | 14 | 12 | **11** | 13 | 14 | 12 | 10 | 10 | 9 | 9 | 9 | 8 | 8 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 3 | 5 | 5 | 10 | 15 | 16 | 19 | 20 | 19 | 17 | **15** | 18 | 20 | 17 | 14 | 14 | 13 | 13 | 12 | 12 | 11 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/ | 3 | 5 | 2 | 4 | 8 | 9 | 11 | 12 | 11 | 10 | **9** | 8 | 6 | 5 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| | | | | | | | | | | | Debt service-to-revenue ratio | | | | | | | | | | |
| **Baseline** | 2 | 1 | 1 | 2 | 3 | 4 | 4 | 5 | 5 | 4 | **4** | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| **A. Alternative Scenarios** | | | | | | | | | | | | | | | | | | | | | |
| A1. Key variables at their historical averages in 2008-2028 1/ | 2 | 2 | 2 | 3 | 5 | 6 | 8 | 9 | 9 | 8 | **7** | 9 | 7 | 6 | 4 | 4 | 3 | 2 | 1 | 0 | 0 |
| A2. New public sector loans on less favorable terms in 2008-2028 2/ | 2 | 2 | 1 | 2 | 4 | 5 | 5 | 5 | 4 | 3 | **3** | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| **B. Bound Tests** | | | | | | | | | | | | | | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2009-2010 | 2 | 2 | 1 | 2 | 4 | 4 | 5 | 5 | 5 | 5 | **4** | 4 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| B2. Export value growth at historical average minus one standard deviation in 2009-2010 3/ | 2 | 2 | 1 | 2 | 3 | 4 | 5 | 5 | 5 | 4 | **4** | 3 | 3 | 3 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2009-2010 | 2 | 2 | 2 | 3 | 5 | 6 | 7 | 8 | 8 | 7 | **6** | 5 | 4 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2009-2010 4/ | 2 | 2 | 2 | 3 | 5 | 5 | 6 | 6 | 6 | 5 | **4** | 6 | 6 | 6 | 5 | 5 | 5 | 4 | 4 | 4 | 4 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 2 | 2 | 2 | 5 | 7 | 7 | 8 | 9 | 9 | 8 | **7** | 8 | 9 | 8 | 7 | 7 | 7 | 7 | 7 | 6 | 6 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2009 5/ | 2 | 2 | 1 | 2 | 4 | 5 | 6 | 7 | 6 | 6 | **5** | 4 | 4 | 3 | 2 | 2 | 3 | 3 | 3 | 3 | 3 |
| *Memorandum item:* | | | | | | | | | | | | | | | | | | | | | |
| Grant element assumed on residual financing (i.e., financing required above baseline) 6/ | 56 | 56 | 56 | 56 | 56 | 56 | 56 | 56 | 56 | 56 | **56** | 56 | 56 | 56 | 56 | 56 | 56 | 56 | 56 | 56 | 56 |

Sources: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3.Burundi: Public Sector Debt Sustainability Framework, Baseline Scenario, 2005-2028
(In percent of GDP, unless otherwise indicated)

| | Actual | | | | | Estimate | Projections | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2005 | 2006 | 2007 | Average | Standard Deviation | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2008-13 Average | 2018 | 2028 | 2014-28 Average |
| **Public sector debt 1/** | 169.8 | 154.8 | 148.8 | | | 122.0 | 26.3 | 27.0 | 27.3 | 26.9 | 26.5 | | 22.7 | 16.0 | |
| o/w foreign-currency denominated | 169.8 | 154.8 | 148.8 | | | 122.0 | 26.3 | 27.0 | 27.3 | 26.9 | 26.5 | | 22.7 | 16.0 | |
| Change in public sector debt | -52.0 | -15.0 | -6.0 | | | -26.8 | -95.7 | 0.7 | 0.3 | -0.4 | -0.4 | | -1.4 | -0.5 | |
| Identified debt-creating flows | -48.6 | -16.1 | -2.7 | | | -29.7 | -80.6 | 1.8 | 2.5 | 3.2 | 3.0 | | -2.5 | 0.4 | |
| Primary deficit | 0.5 | -0.1 | -3.2 | 0.7 | 2.2 | -1.4 | -59.7 | 2.7 | 3.5 | 4.3 | 4.3 | -7.7 | -0.9 | 1.5 | 1.4 |
| Revenue and grants | 31.7 | 36.8 | 39.5 | | | 44.6 | 103.0 | 39.8 | 37.5 | 36.2 | 36.2 | | 28.9 | 25.2 | |
| *of which: grants* | 11.7 | 17.9 | 20.9 | | | 25.5 | 83.7 | 20.4 | 18.1 | 16.6 | 16.4 | | 8.1 | 4.2 | |
| Primary (noninterest) expenditure | 32.2 | 36.6 | 36.4 | | | 43.2 | 43.2 | 42.5 | 41.0 | 40.5 | 40.4 | | 27.9 | 26.7 | |
| Automatic debt dynamics | -47.9 | -13.0 | 1.6 | | | -28.3 | -20.9 | -0.9 | -1.0 | -1.1 | -1.3 | | -1.5 | -1.0 | |
| Contribution from interest rate/growth differential | -4.0 | -11.8 | -7.2 | | | -7.8 | -6.0 | -0.9 | -1.1 | -1.2 | -1.3 | | -1.4 | -1.0 | |
| *of which: contribution from average real interest rate* | -2.0 | -3.5 | -1.8 | | | -1.3 | -0.8 | 0.4 | 0.2 | 0.1 | 0.0 | | -0.2 | -0.2 | |
| *of which: contribution from real GDP growth* | -2.0 | -8.3 | -5.3 | | | -6.4 | -5.2 | -1.2 | -1.3 | -1.3 | -1.3 | | -1.2 | -0.8 | |
| Contribution from real exchange rate depreciation | -43.9 | -1.2 | 8.8 | | | -20.5 | -14.8 | 0.0 | 0.0 | 0.1 | 0.1 | | ... | ... | |
| Other identified debt-creating flows | -1.1 | -3.0 | -1.2 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Privatization receipts (negative) | -0.3 | -1.1 | -0.8 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Recognition of implicit or contingent liabilities | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Debt relief (HIPC and other) | -0.8 | -1.9 | -0.3 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Other (specify, e.g. bank recapitalization) | 0.0 | 0.0 | 0.0 | | | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | | 0.0 | 0.0 | |
| Residual, including asset changes | -3.5 | 1.1 | -3.3 | | | 2.9 | -15.0 | -1.2 | -2.2 | -3.6 | -3.4 | | 1.1 | -0.9 | |
| **Other Sustainability Indicators** | | | | | | | | | | | | | | | |
| **PV of public sector debt** | 0.0 | 0.0 | 9.3 | | | 9.3 | 9.7 | 11.3 | 12.5 | 12.4 | 12.2 | | 10.0 | 7.5 | |
| o/w foreign-currency denominated | 0.0 | 0.0 | 9.3 | | | 9.3 | 9.7 | 11.3 | 12.5 | 12.4 | 12.2 | | 10.0 | 7.5 | |
| o/w external | ... | ... | 9.3 | | | 9.3 | 9.7 | 11.3 | 12.5 | 12.4 | 12.2 | | 10.0 | 7.5 | |
| PV of contingent liabilities (not included in public sector debt) | ... | ... | ... | | | ... | ... | ... | ... | ... | ... | | ... | ... | |
| Gross financing need 2/ | 7.1 | 4.7 | -1.7 | | | -0.8 | 0.5 | -58.8 | 4.2 | 4.7 | 5.7 | | 1.8 | 1.7 | |
| PV of public sector debt-to-revenue and grants ratio (in percent) | 0.0 | 0.0 | 23.4 | | | 20.8 | 9.4 | 28.3 | 33.4 | 34.3 | 33.9 | | 34.7 | 29.7 | |
| PV of public sector debt-to-revenue ratio (in percent) | 0.0 | 0.1 | 49.7 | | | 48.8 | 50.5 | 58.0 | 64.3 | 63.5 | 61.9 | | 48.3 | 35.6 | |
| o/w external 3/ | … | … | 49.6 | | | 48.7 | 50.4 | 57.9 | 64.3 | 63.5 | 61.9 | | 48.3 | 35.6 | |
| Debt service-to-revenue and grants ratio (in percent) 4/ | 12.1 | -4.1 | 6.0 | | | 4.2 | 2.2 | 4.0 | 3.5 | 4.1 | 4.2 | | 3.7 | 1.8 | |
| Debt service-to-revenue ratio (in percent) 4/ | 22.8 | 8.0 | 11.3 | | | 8.4 | 7.2 | 4.5 | 3.8 | 3.3 | 2.5 | | 1.0 | 0.5 | |
| Primary deficit that stabilizes the debt-to-GDP ratio | 52.5 | 14.9 | 2.9 | | | 25.4 | 35.9 | 2.0 | 3.2 | 4.7 | 4.7 | | 0.4 | 1.9 | |
| **Key macroeconomic and fiscal assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 0.9 | 5.1 | 3.6 | 2.3 | 2.6 | 4.5 | 4.5 | 5.0 | 5.0 | 5.0 | 5.0 | 4.8 | 5.0 | 5.0 | 5.0 |
| Average nominal interest rate on forex debt (in percent) | 1.3 | 0.1 | 0.3 | 0.8 | 0.6 | 0.0 | 0.1 | 0.5 | 0.6 | 0.6 | 0.6 | 0.4 | 0.7 | 0.7 | 0.7 |
| Real exchange rate depreciation (in percent, + indicates depreciation) | -22.2 | -5.0 | 12.0 | 12.2 | 22.4 | -7.5 | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| Inflation rate (GDP deflator, in percent) | 19.4 | 9.2 | 2.4 | -1.7 | 10.6 | 14.8 | 15.8 | 7.2 | 1.3 | 1.7 | 1.7 | 7.1 | 2.7 | 2.4 | 2.4 |
| Growth of real primary spending (deflated by GDP deflator, in percent) | -4.2 | 19.7 | 2.8 | 2.4 | 7.3 | 24.0 | 4.7 | 3.2 | 1.3 | 3.7 | 4.7 | 6.9 | -5.1 | 5.1 | 2.2 |
| Grant element of new external borrowing (in percent) | ... | ... | ... | … | … | 52.5 | 48.7 | 48.6 | 48.5 | 58.8 | 58.8 | 52.6 | 58.8 | 58.8 | ... |

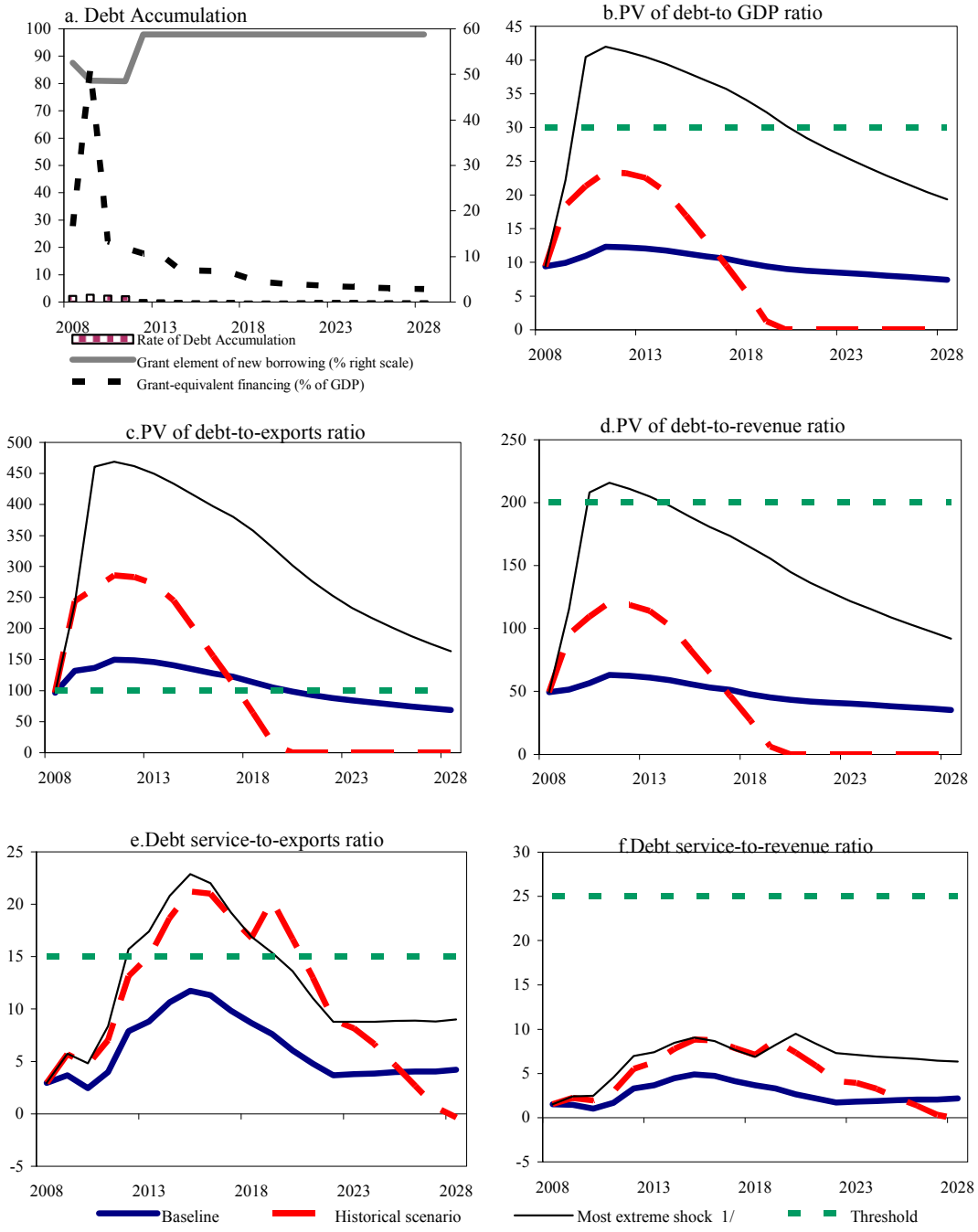
Sources: Burundi authorities; and Fund staff estimates and projections.
1/ Central government on a gross basis.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

Table 4. Burundi: Sensitivity Analysis for Key Indicators of Public Debt 2008-2028

| | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2008 | 2009 | 2010 | 2011 | 2012 | 2013 | 2018 | 2028 |
| **PV of Debt-to-GDP Ratio** | | | | | | | | |
| **Baseline** | 9 | 10 | 11 | 13 | 12 | 12 | 10 | 7 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 9 | 39 | 40 | 40 | 39 | 37 | 32 | 20 |
| A2. Primary balance is unchanged from 2008 | 9 | 39 | 36 | 35 | 31 | 28 | 17 | ... |
| A3. Permanently lower GDP growth 1/ | 9 | 39 | 12 | 13 | 13 | 13 | 13 | 18 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010 | 9 | 11 | 14 | 16 | 17 | 18 | 19 | 23 |
| B2. Primary balance is at historical average minus one standard deviations in 2009-2010 | 9 | 39 | 40 | 41 | 40 | 39 | 33 | 20 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 9 | 40 | 42 | 43 | 43 | 43 | 40 | 31 |
| B4. One-time 30 percent real depreciation in 2009 | 9 | 45 | 43 | 41 | 38 | 35 | 23 | 12 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2009 | 9 | 14 | 16 | 17 | 17 | 17 | 14 | 9 |
| **PV of Debt-to-Revenue Ratio 2/** | | | | | | | | |
| **Baseline** | 21 | 9 | 28 | 33 | 34 | 34 | 35 | 30 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 21 | 37 | 98 | 104 | 102 | 97 | 104 | 71 |
| A2. Primary balance is unchanged from 2008 | 21 | 38 | 91 | 92 | 87 | 77 | 58 | ... |
| A3. Permanently lower GDP growth 1/ | 21 | 38 | 29 | 34 | 36 | 36 | 45 | 71 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010 | 21 | 10 | 33 | 41 | 45 | 47 | 65 | 88 |
| B2. Primary balance is at historical average minus one standard deviations in 2009-2010 | 21 | 38 | 101 | 109 | 111 | 108 | 115 | 78 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 21 | 38 | 102 | 112 | 116 | 115 | 136 | 120 |
| B4. One-time 30 percent real depreciation in 2009 | 21 | 44 | 107 | 109 | 105 | 97 | 80 | 49 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2009 | 21 | 14 | 40 | 45 | 46 | 46 | 48 | 37 |
| **Debt Service-to-Revenue Ratio 2/** | | | | | | | | |
| **Baseline** | 4 | 2 | 4 | 4 | 4 | 4 | 4 | 2 |
| **A. Alternative scenarios** | | | | | | | | |
| A1. Real GDP growth and primary balance are at historical averages | 4 | 2 | 5 | 4 | 5 | 5 | 5 | 6 |
| A2. Primary balance is unchanged from 2008 | 4 | 2 | 5 | 4 | 5 | 5 | 4 | 2 |
| A3. Permanently lower GDP growth 1/ | 4 | 2 | 4 | 4 | 4 | 4 | 4 | 3 |
| **B. Bound tests** | | | | | | | | |
| B1. Real GDP growth is at historical average minus one standard deviations in 2009-2010 | 4 | 2 | 4 | 4 | 4 | 4 | 4 | 4 |
| B2. Primary balance is at historical average minus one standard deviations in 2009-2010 | 4 | 2 | 5 | 4 | 5 | 5 | 4 | 6 |
| B3. Combination of B1-B2 using one half standard deviation shocks | 4 | 2 | 5 | 5 | 5 | 5 | 5 | 7 |
| B4. One-time 30 percent real depreciation in 2009 | 4 | 2 | 4 | 4 | 5 | 5 | 5 | 3 |
| B5. 10 percent of GDP increase in other debt-creating flows in 2009 | 4 | 2 | 4 | 4 | 4 | 4 | 4 | 2 |

Sources: Burundi authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the length of the projection period.
2/ Revenues are defined inclusive of grants.

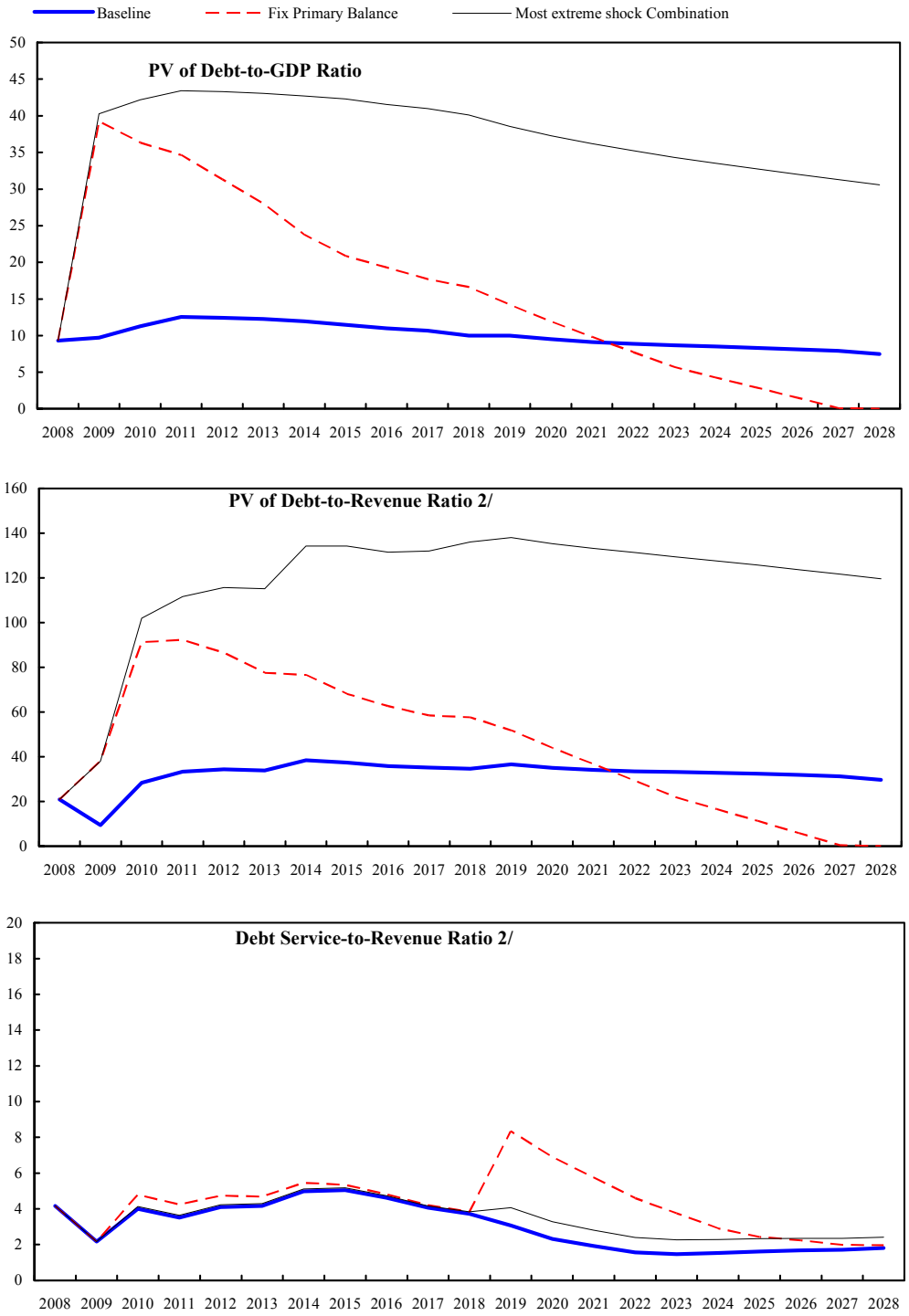
Figure 1. Burundi: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2008-2028 1/




Sources: Staff projections and simulations.

1/ The most extreme stress test is the test that yields the highest ratio in 2018. In figure b. it corresponds to a Combination shock; in c. to a Combination shock; in d. to a Combination shock; in e. to a Exports shock and in picture f. to a Combination shock.

Figure 2.Burundi: Indicators of Public Debt Under Alternative Scenarios, 2008-2028 1/




Sources: Country authorities; and Fund staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2018.
2/ Revenues are defined inclusive of grants.